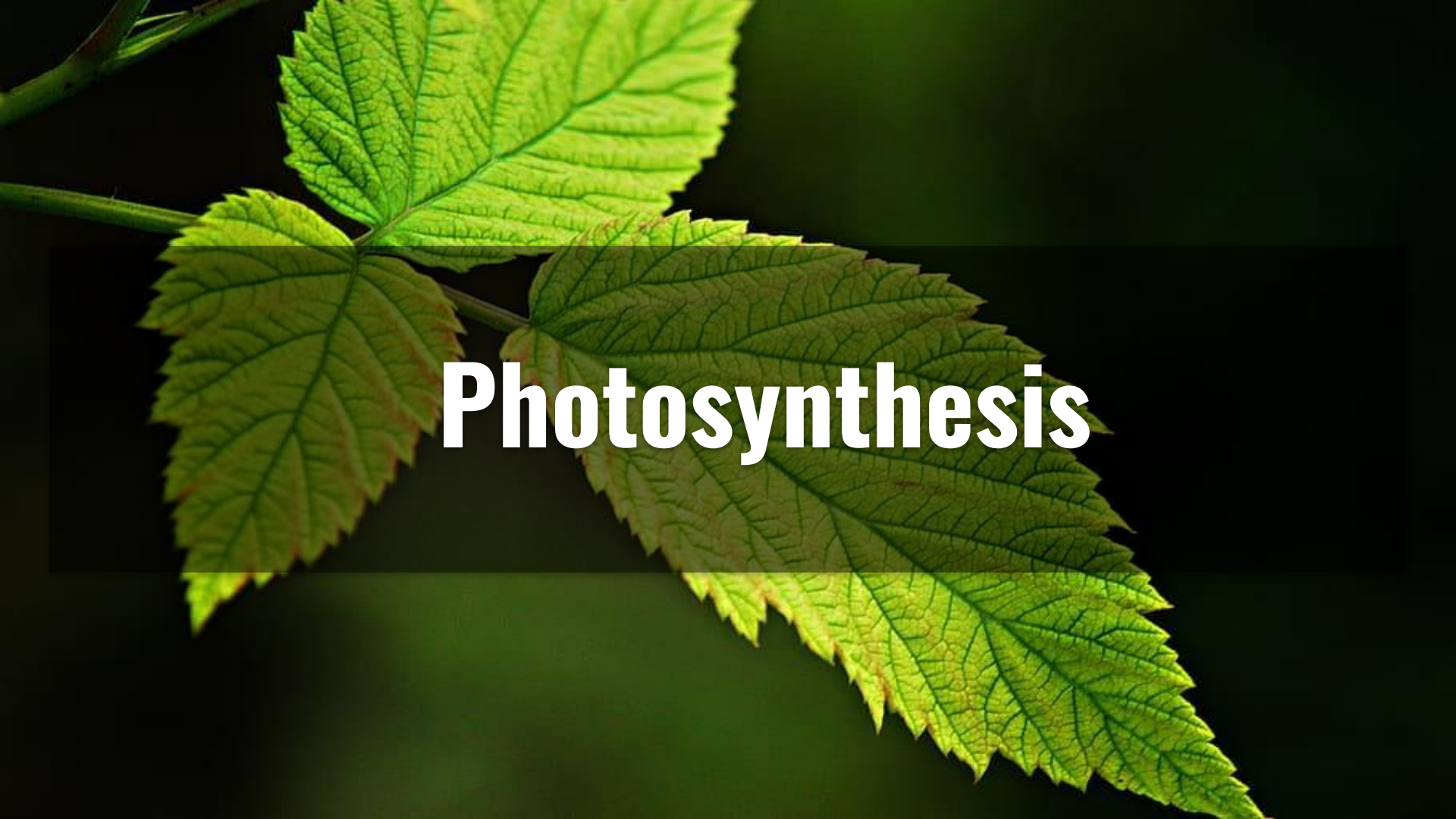

Photosynthesis

How Organisms Obtain Energy

- **Autotrophs** (AKA Producers): make their own food by building organic molecules
- **Heterotrophs** (AKA Consumers): Obtain energy from foods they consume





Plant Anatomy Basics

- **Roots:** soak up water and nutrients from the soil, may also serve to store extra sugars
- **Stems** : bring water and nutrients to different parts of the plant via capillary action
- **Leaves** : exchange gasses and water between the plant and air through the tiny openings called **stomata**
- **Transpiration** : evaporation through the stomata of a plant



What do plants need?

- Need to collect **light energy** →leaves
- Need to store light energy → chloroplasts
- Need to get the **building block** atoms from the environment (C, H, O, N, P, S, K, Mg) → stomata (CO2 and O2) and roots (water, nutrients)
- Need to produce the **organic molecules** necessary for growth
    - Carbs, lipids, proteins, and nucleic acids!

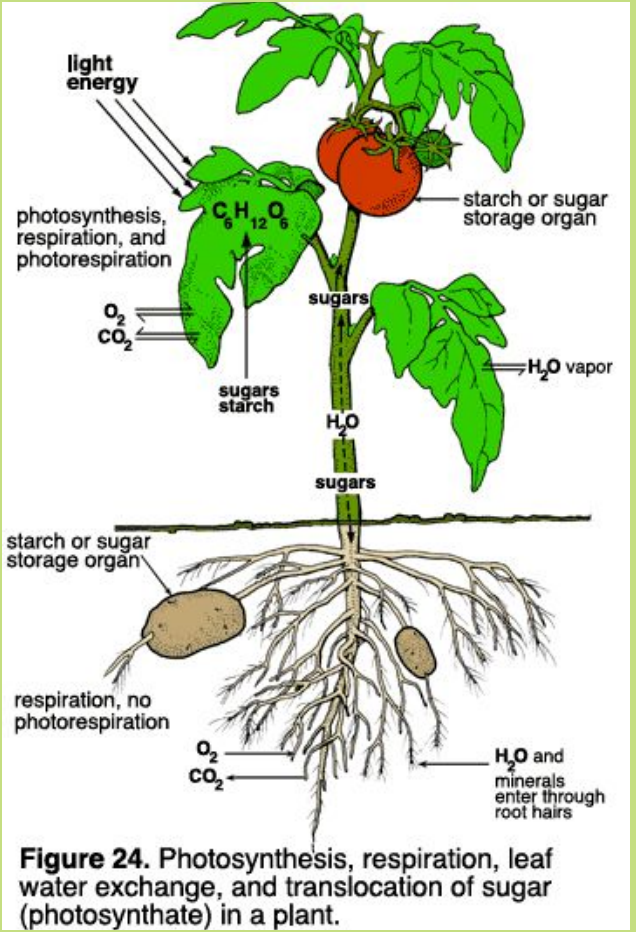
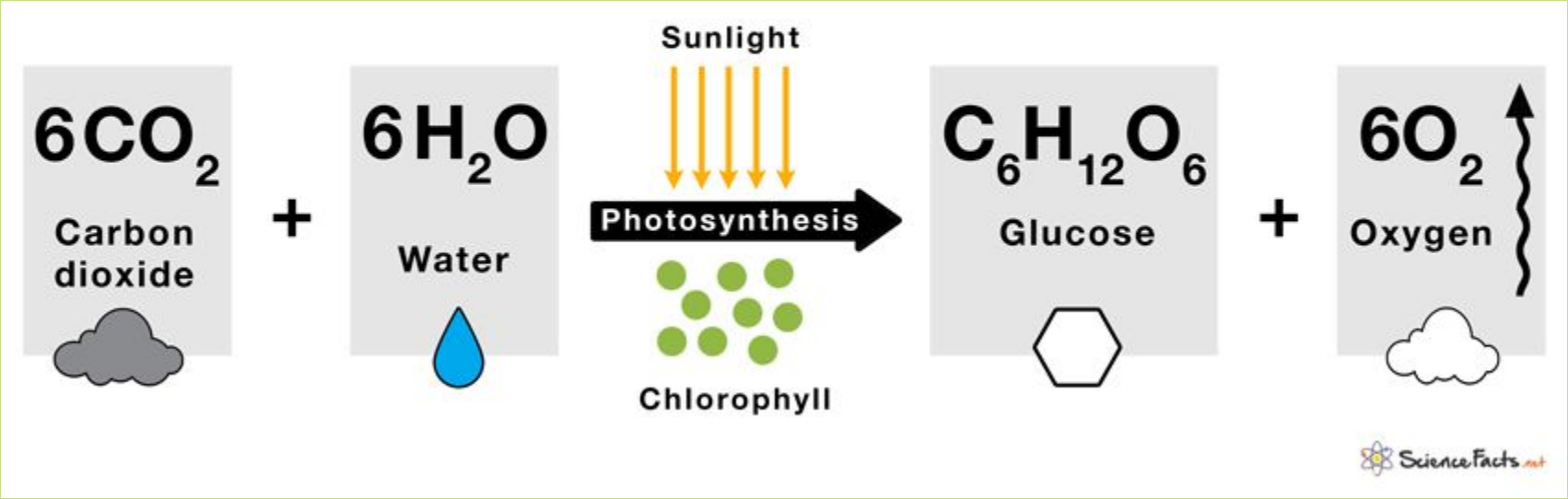


Figure 24. Photosynthesis, respiration, leaf water exchange, and translocation of sugar (photosynthate) in a plant.

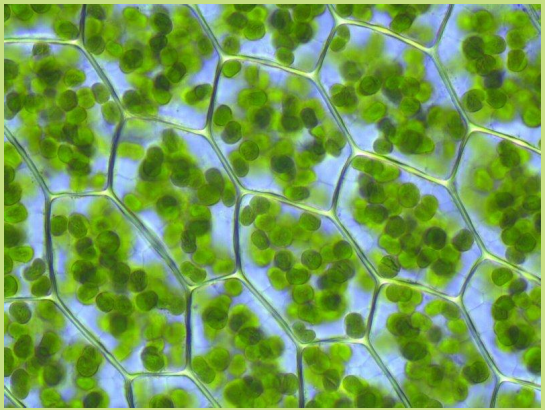
Photosynthesis Equation

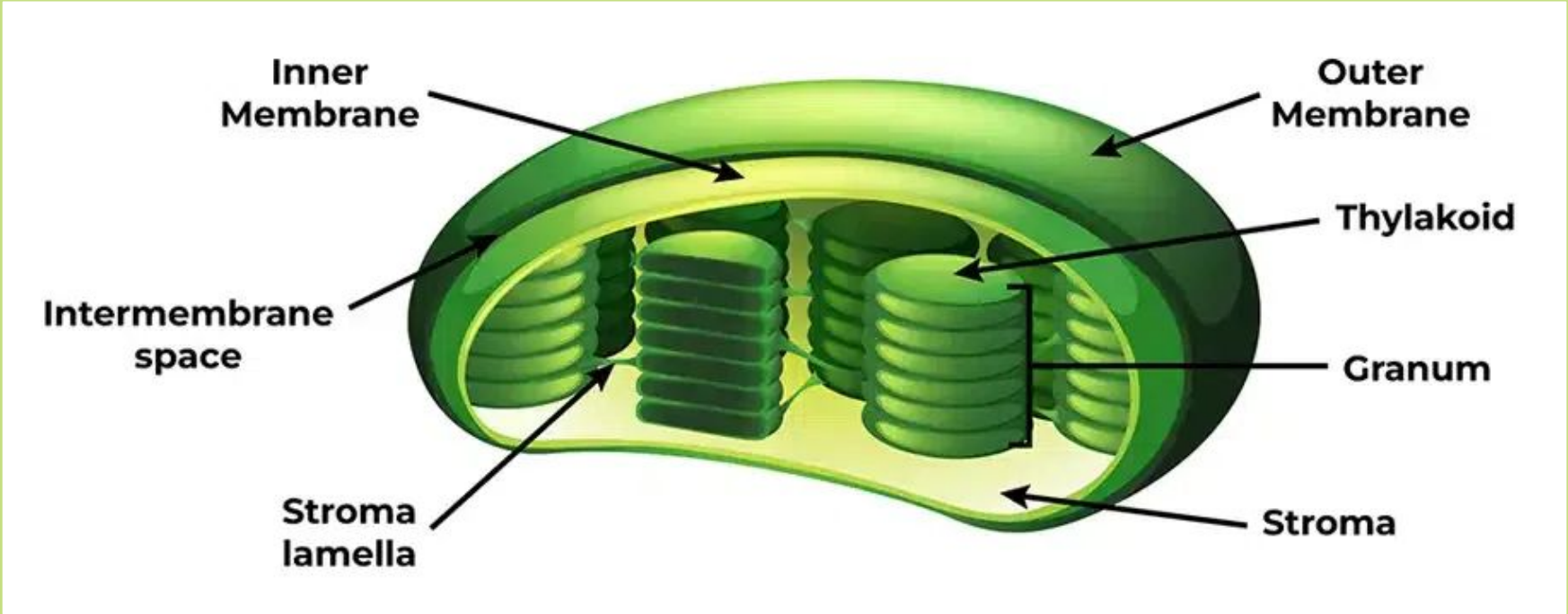
How is this equation similar to the equation for cellular respiration?



The Chloroplast

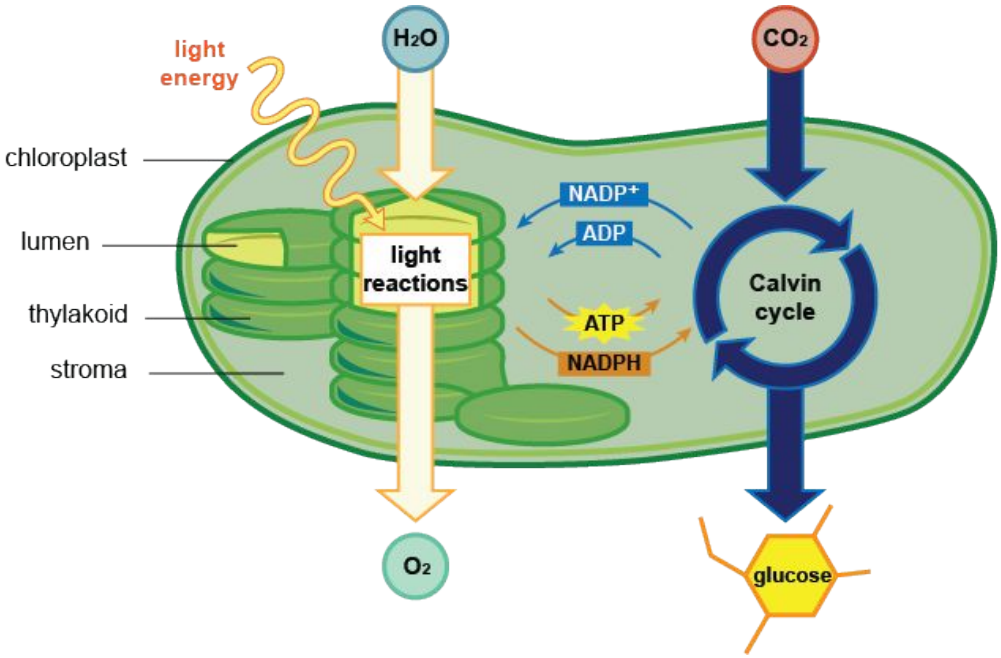
- Photosynthesis occurs inside the **chloroplast**
- Two main parts:
  - **Grana:** Stacks of thylakoids
  - **Stroma** : Fluid inside of the chloroplast
- **Thylakoid membrane** contains:
  - Chlorophyll molecules
  - Electron Transport Chain
  - ATP Synthase
- Double membrane, ribosomes, and DNA just like mitochondria!




Inner Membrane
Outer Membrane
Thylakoid
Intermembrane space
Granum
Stroma lamella
Stroma

Photosynthesis Overview

- Two sets of reactions happen inside the chloroplast
- **Light reactions** (Light dependent)- sunlight to ATP
- **Calvin Cycle** (Light independent)- CO2 to glucose

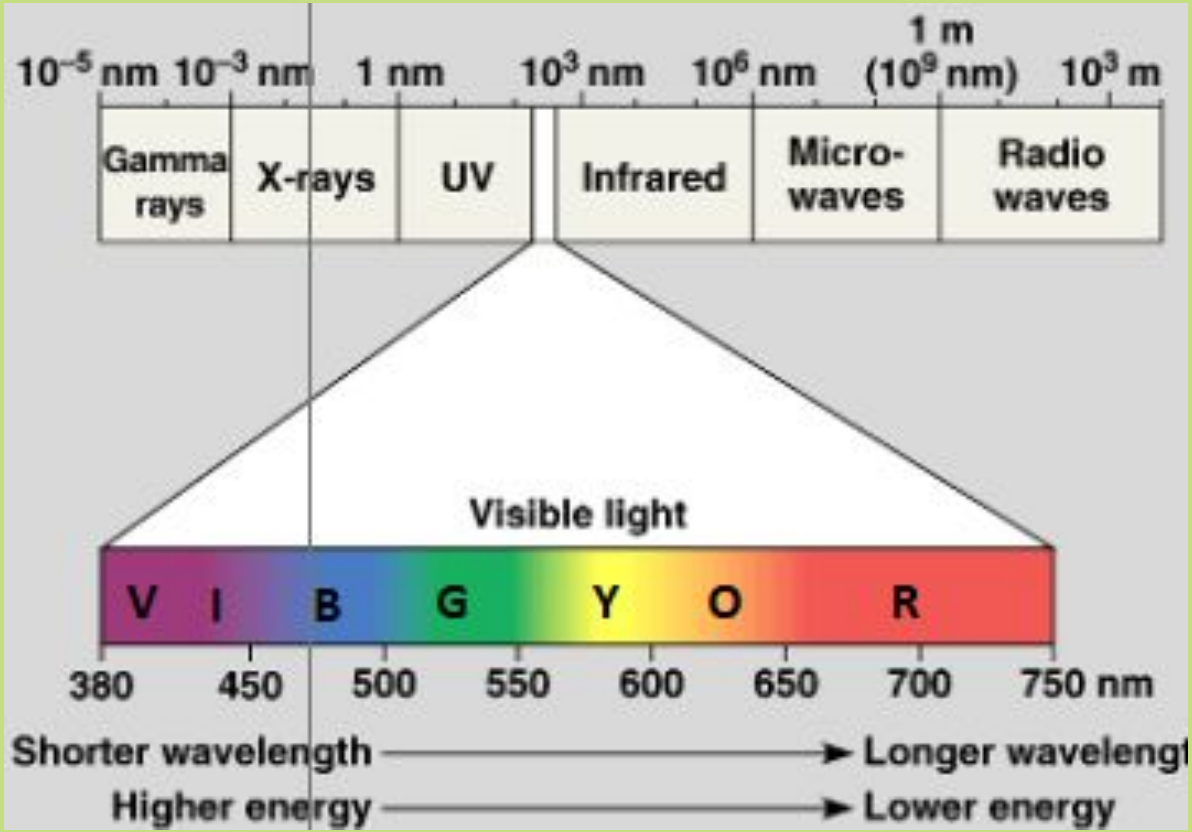


Light Absorption

- **Visible light** is made of different wavelengths
- The wavelength determines the color of the light
- Sunlight is “white” light made up of a mixture of different wavelengths

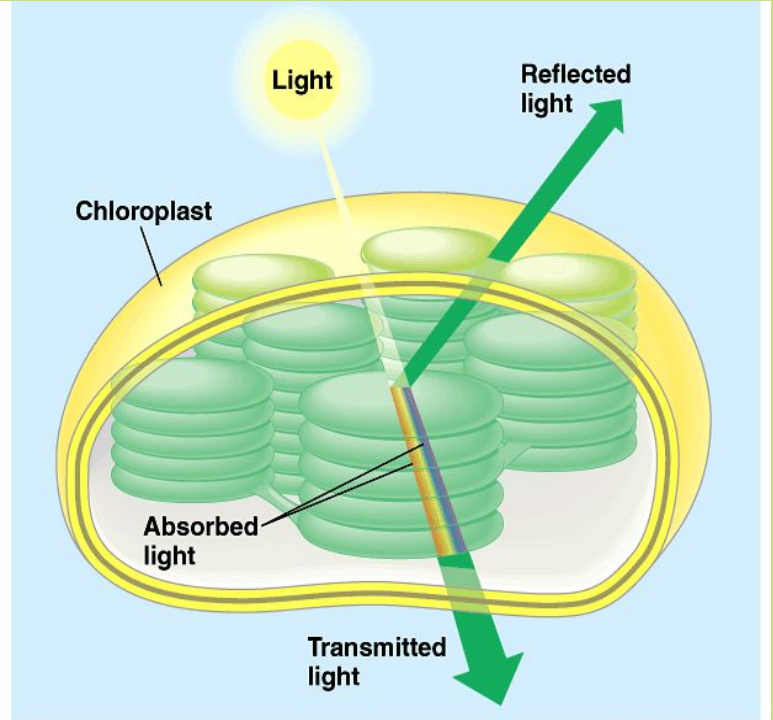


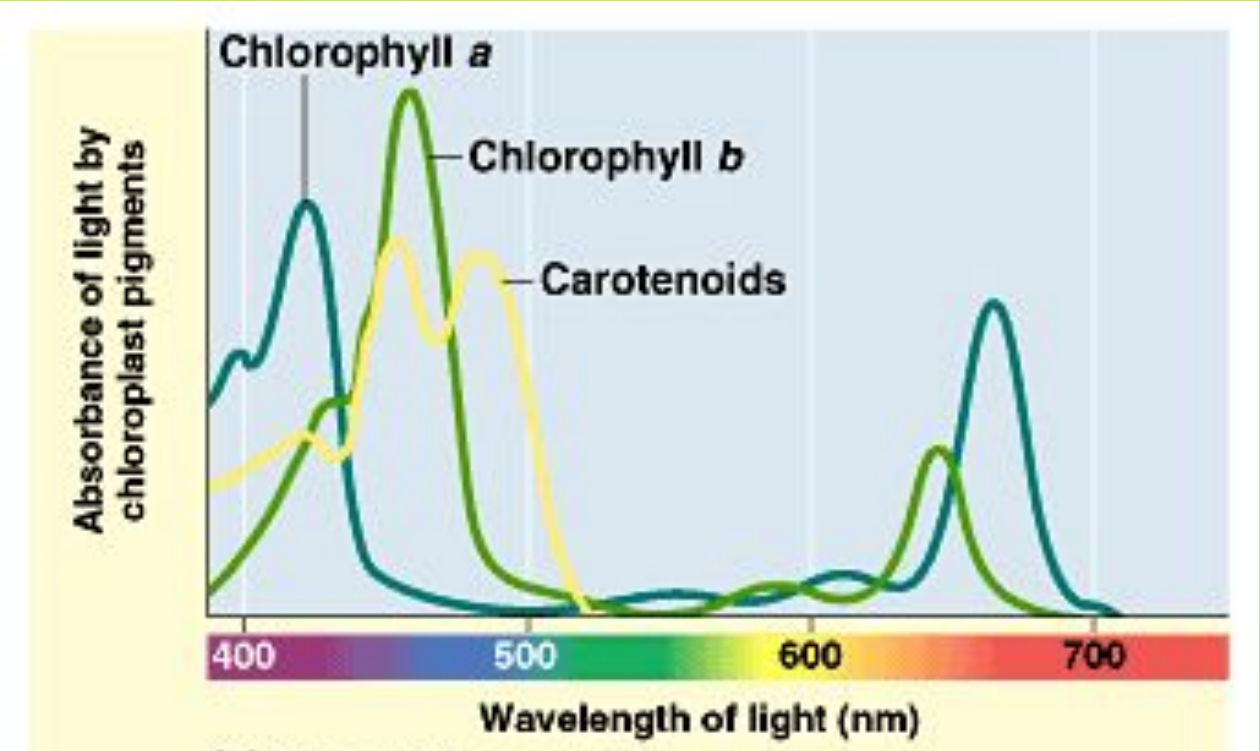
The Spectrum of Color



Pigments of Photosynthesis

- **Chlorophyll** and other pigments embedded in the thylakoid membrane are responsible for light absorption
- **Chlorophyll a** : absorbs best in red and blue wavelengths of light
- **Accessory pigments** help absorb other wavelengths (ex. chlorophyll b, carotenoids, and xanthophylls)

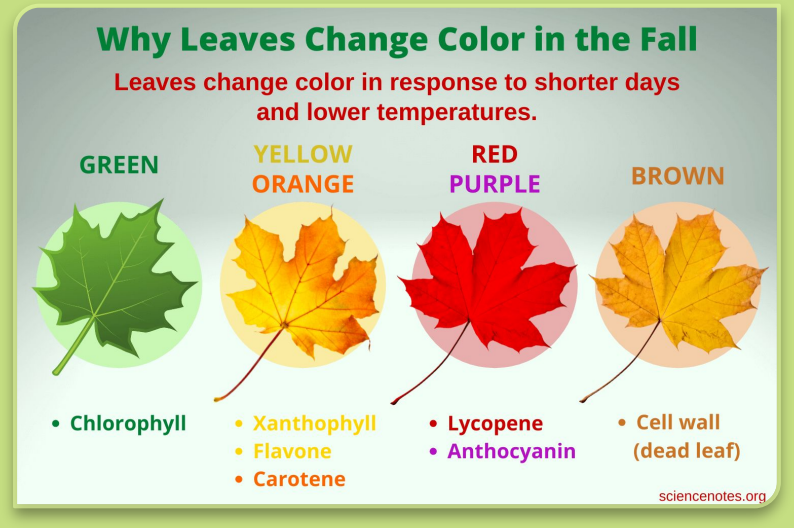




Which color of light do plants ***<u>not</u>*** *absorb?*

Pigments of Photosynthesis

- **Accessory pigments** absorb light energy in other portions of the light spectrum
- Include: chlorophyll b, carotene (red/orange pigment), and xanthophylls
- Why do leaves change in the fall?
  - Temperature drops cause a decrease in chlorophyll A and B production
  - Allows accessory pigments (red, orange, and yellow) to show turning the leaf a different color

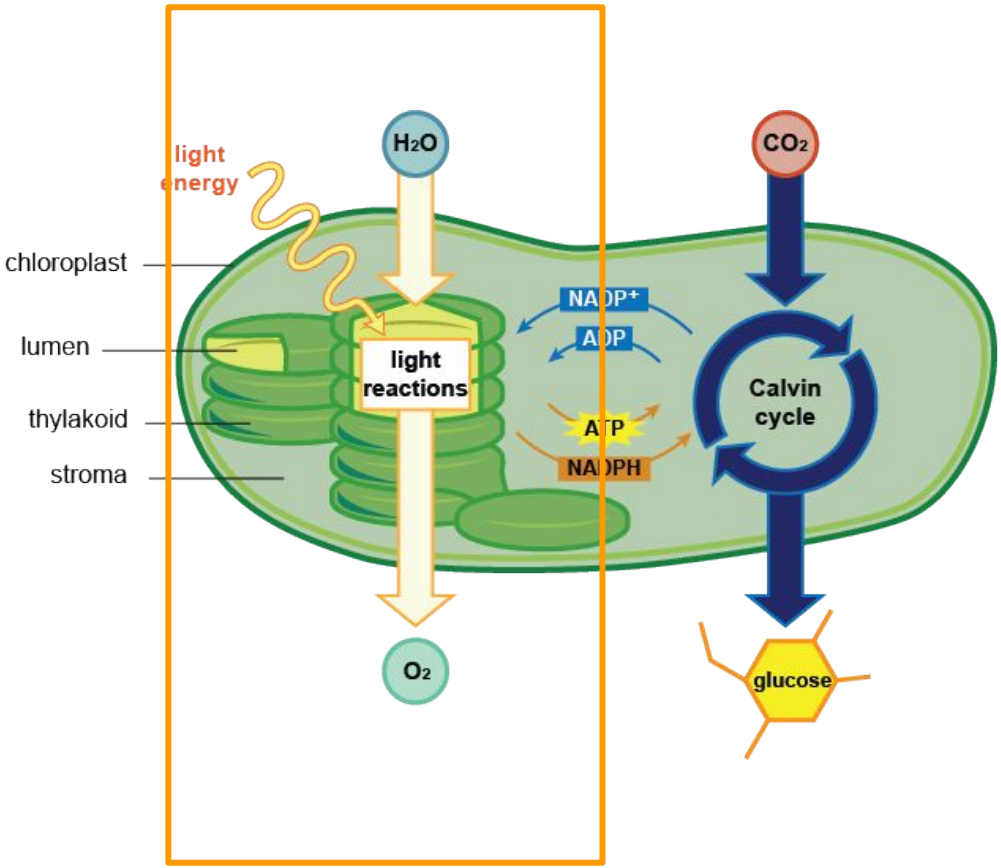




Light Dependent Reactions

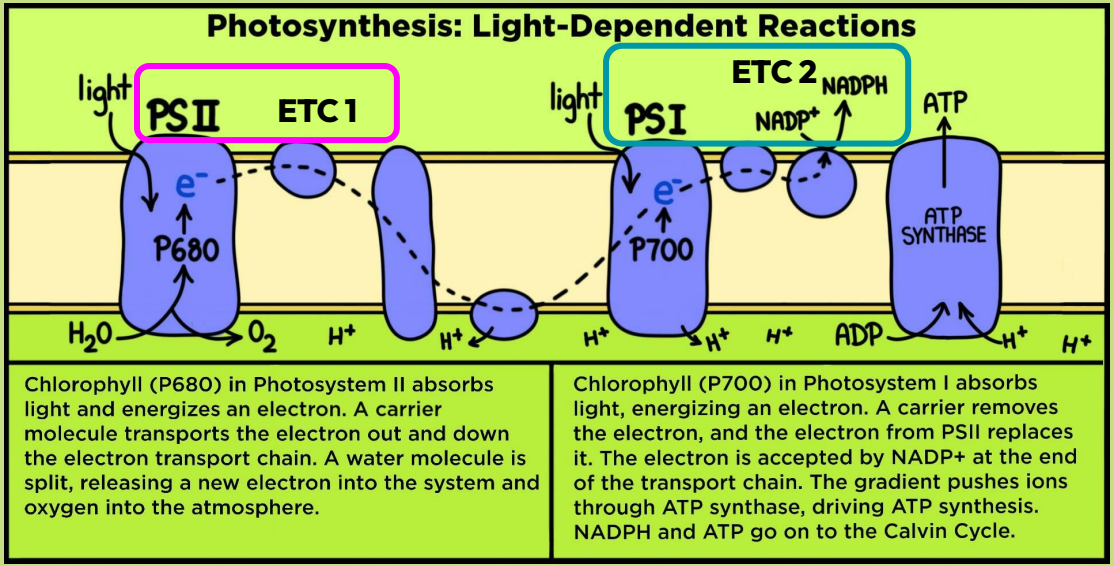
Light Reactions Overview

- Location: **Thylakoid membrane**
- Summary: chlorophyll absorbs light and converts it into chemical energy (ATP and NADPH)
- Water goes in and oxygen comes out as a byproduct
- Two photosystems (I and II) play a part in light reactions
    - **Photosystems** = large complexes of proteins and pigments

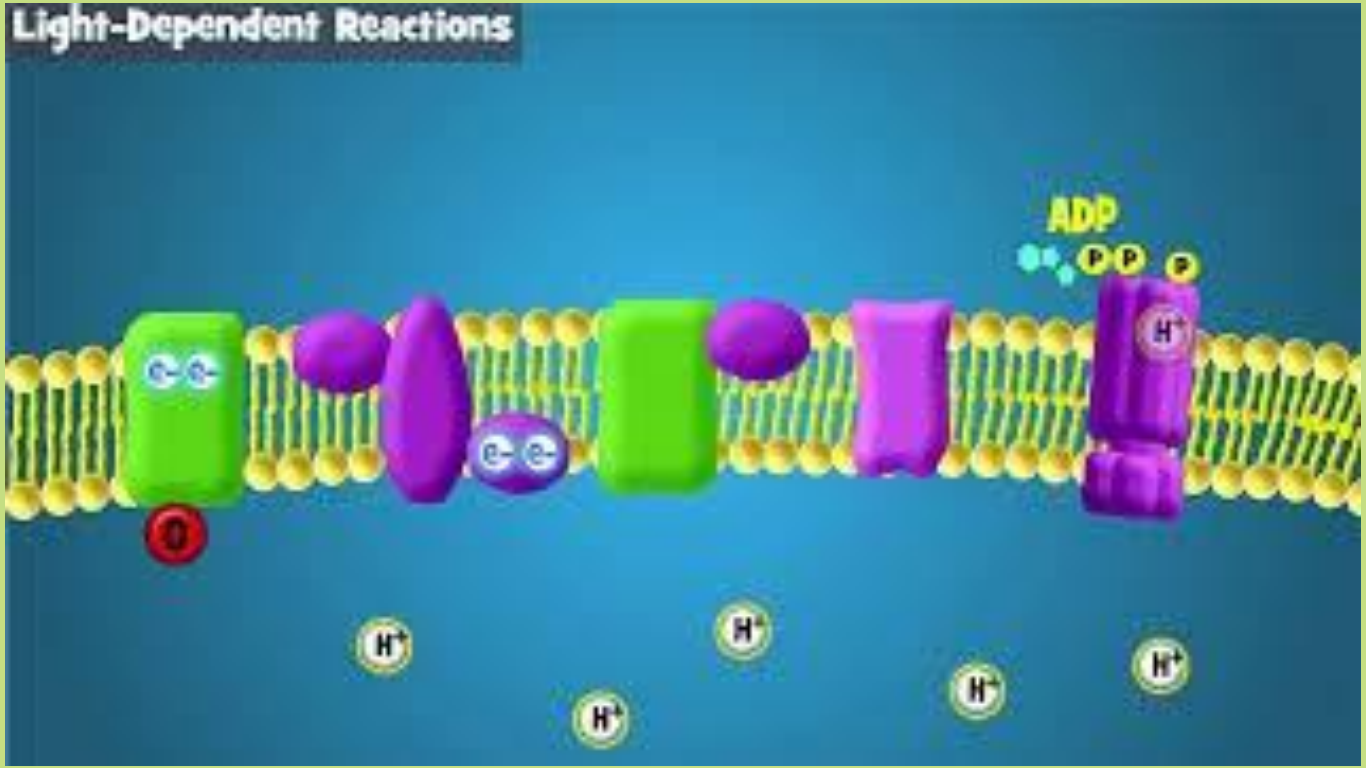


Light Reactions

- Like cellular respiration there is an **ETC** to create a H+ gradient!
- ETC uses light energy to produce ATP and NADPH to power the Calvin cycle
- **ETC 1** pairs with **photosystem II** to make ATP
- **ETC 2** pairs with **photosystem I** to make NADPH



Light Reactions Video

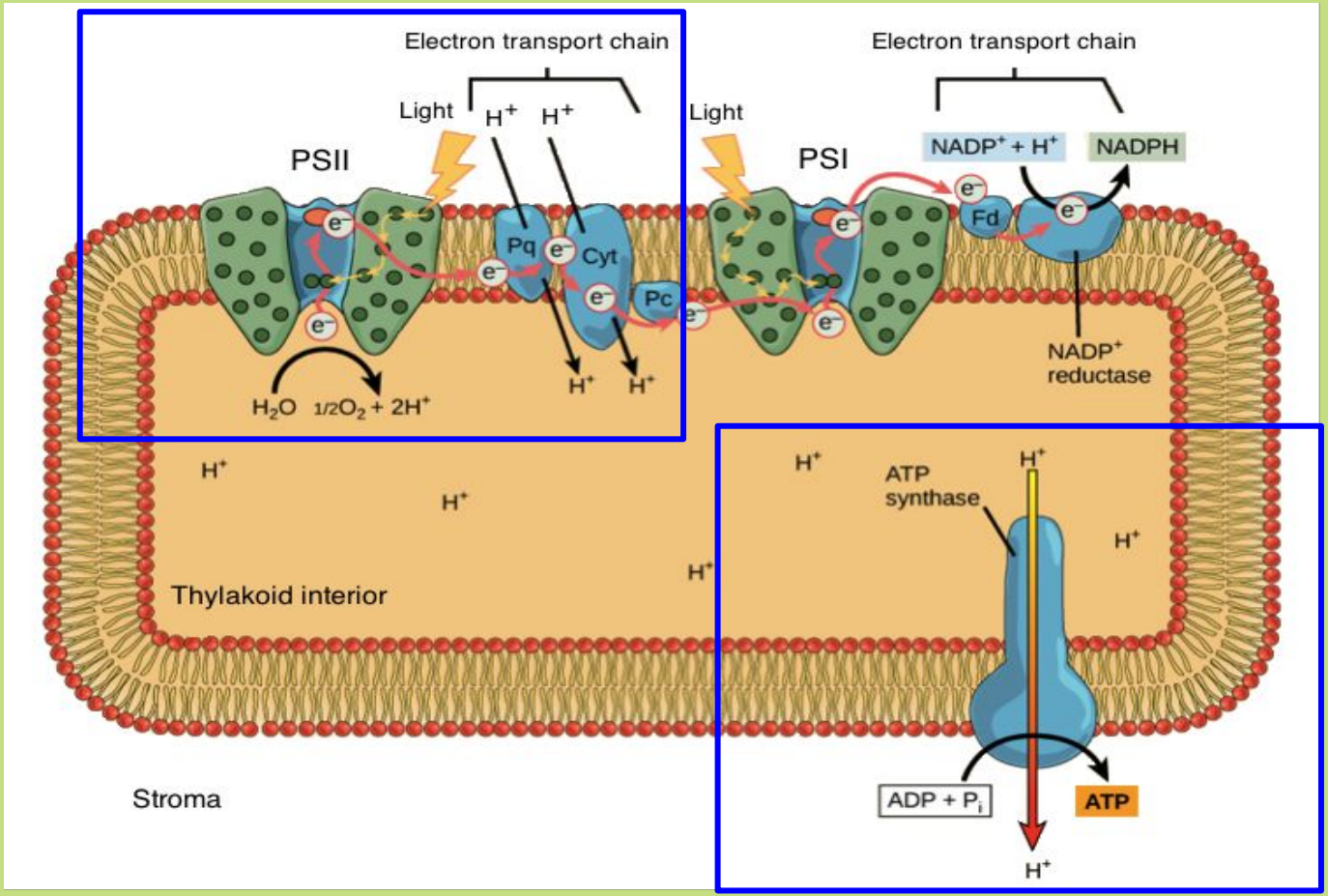


Step 1: Photosystem II/ETC 1

1. Light is absorbed by pigments in PSII and electrons in chlorophyll become excited (high energy)
2. Electrons leave chlorophyll and enter the **ETC**
3. Water is split into H+, e-, and **O_2** (replenishes electrons)
4. Energy from electrons power the ETC and move H+ INTO the thylakoid
5. H+ diffuses back out to the stroma through **ATP synthase** to produce ATP

Where does the oxygen go?

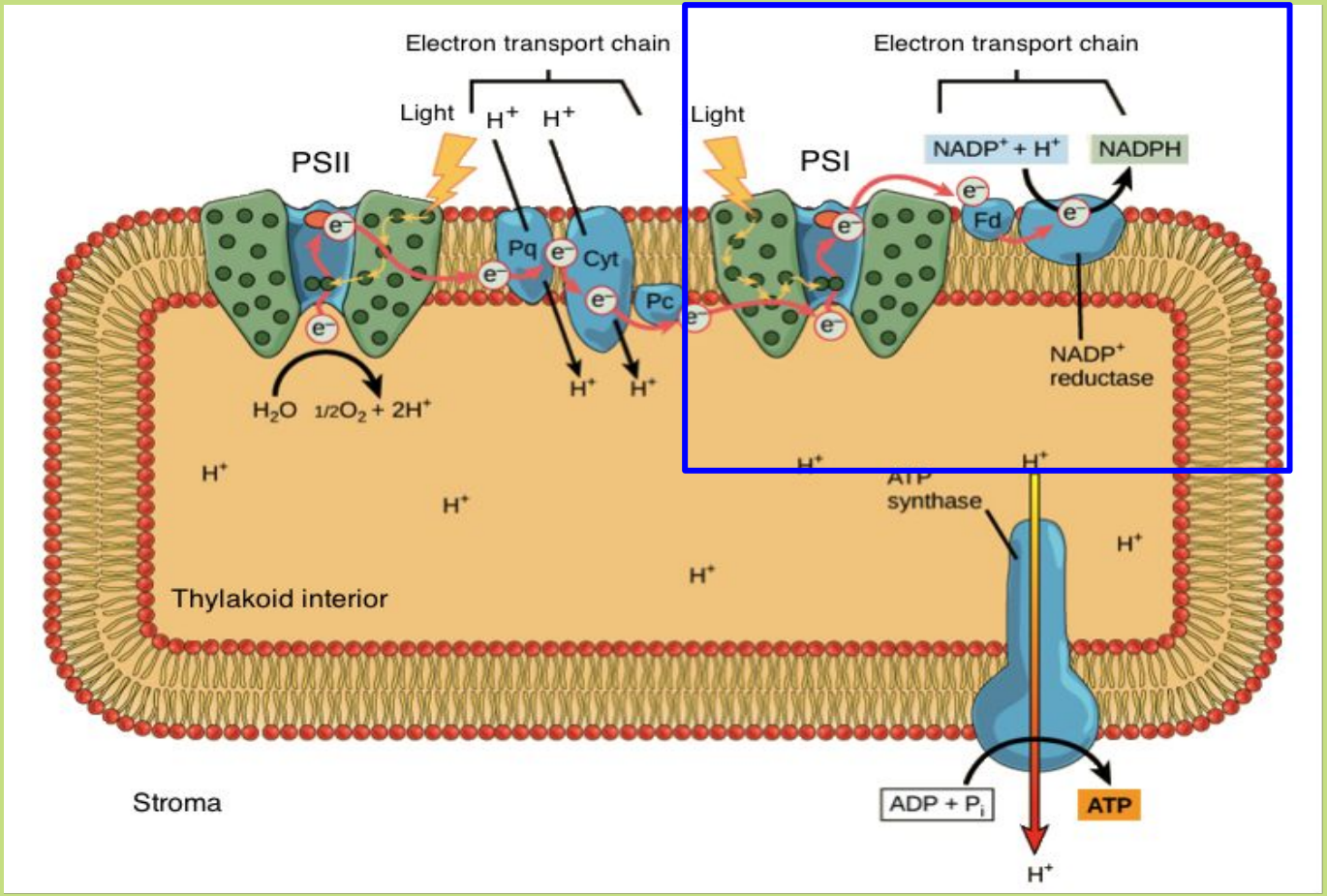
Photosystem II/ETC 1 (Step 1)



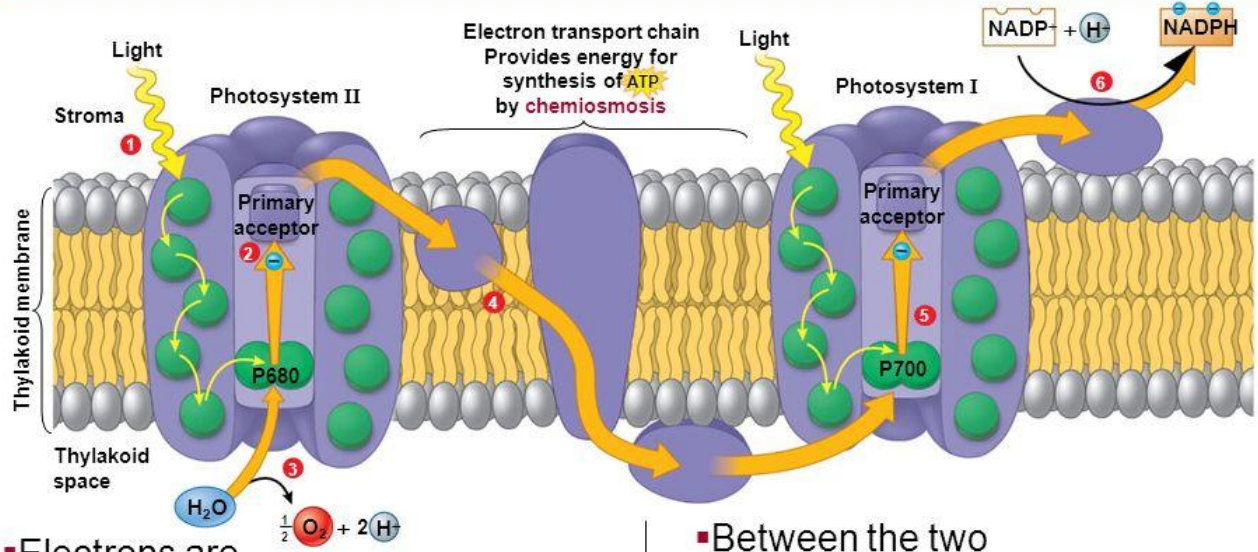
Step 2: Photosystem I/ETC 2

1. Electrons from PSII enter PS I and get re-energized with light absorbed by pigments
2. As e-'s move into ETC 2 an enzyme called **NADP+ reductase** reduces NADP+ into NADPH (accepts electrons)
3. NADPH and ATP are now in the stroma ready to power the light independent reactions (Calvin cycle)

Photosystem I/ETC 2 (Step 2)

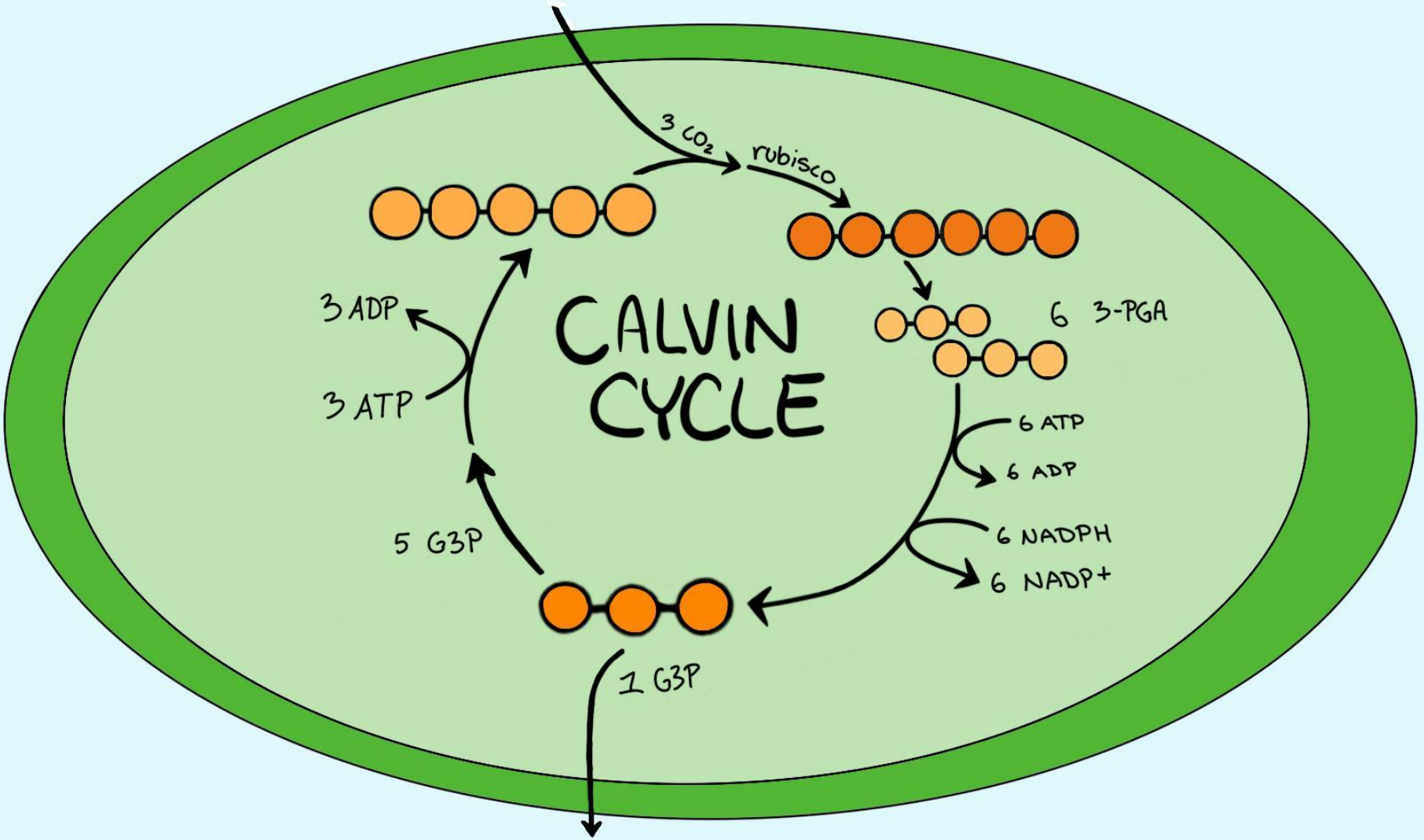


7.8 Two photosystems connected by an electron transport chain generate ATP and NADPH



- Electrons are
  - –removed from water,
  - –passed from PS II to PSI and
  - –accepted by $NADP^+$, reducing it to NADPH.

- Between the two photosystems, the electrons
  - –move down an electron transport chain and
  - –provide energy for the synthesis of ATP.

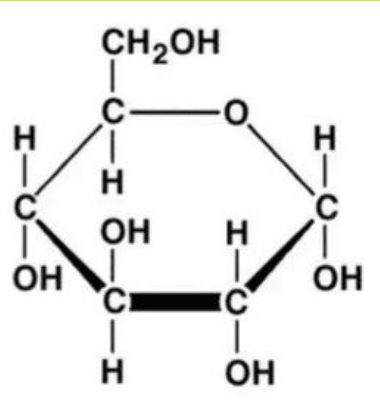

3 CO_2
rubisco
6 3-PGA
CALVIN CYCLE
3 ADP
3 ATP
6 ATP
6 ADP
6 NADPH
6 NADP+
5 G3P
1 G3P

Review: What do plants need?

- Plants need to produce their own organic molecules necessary for growth
- What do we already have?
  - ATP and NADPH (chemical energy) from light rxns
- Plants need to store this chemical energy in a more stable form we will store it into organic molecules (glucose!)
- Where is this sugar stored? (...think back to Unit 1 again)

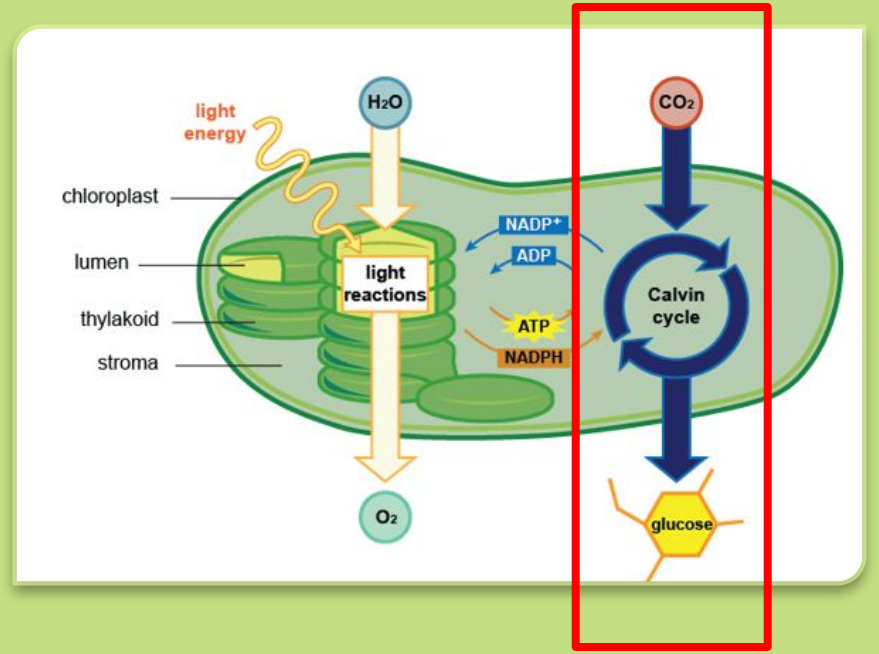
How is Glucose Made?

- Glucose = $C_6H_{12}O_6$
- CO_2 that enters the plant through stomata can be “fixed” (incorporated into organic molecules) and used to build sugars
  - In this process CO_2 is **reduced**
- ATP and NADPH power this reaction



Calvin Cycle Overview

- Location: **stroma** inside chloroplasts
- Powered by energy generated in light reactions
- CO2 has very little chemical energy
  - Calvin cycle converts CO_2 into $C_6H_{12}O_6$ which has a lot of chemical energy
- Synthesizing glucose=**endergonic process** completed by reducing CO_2

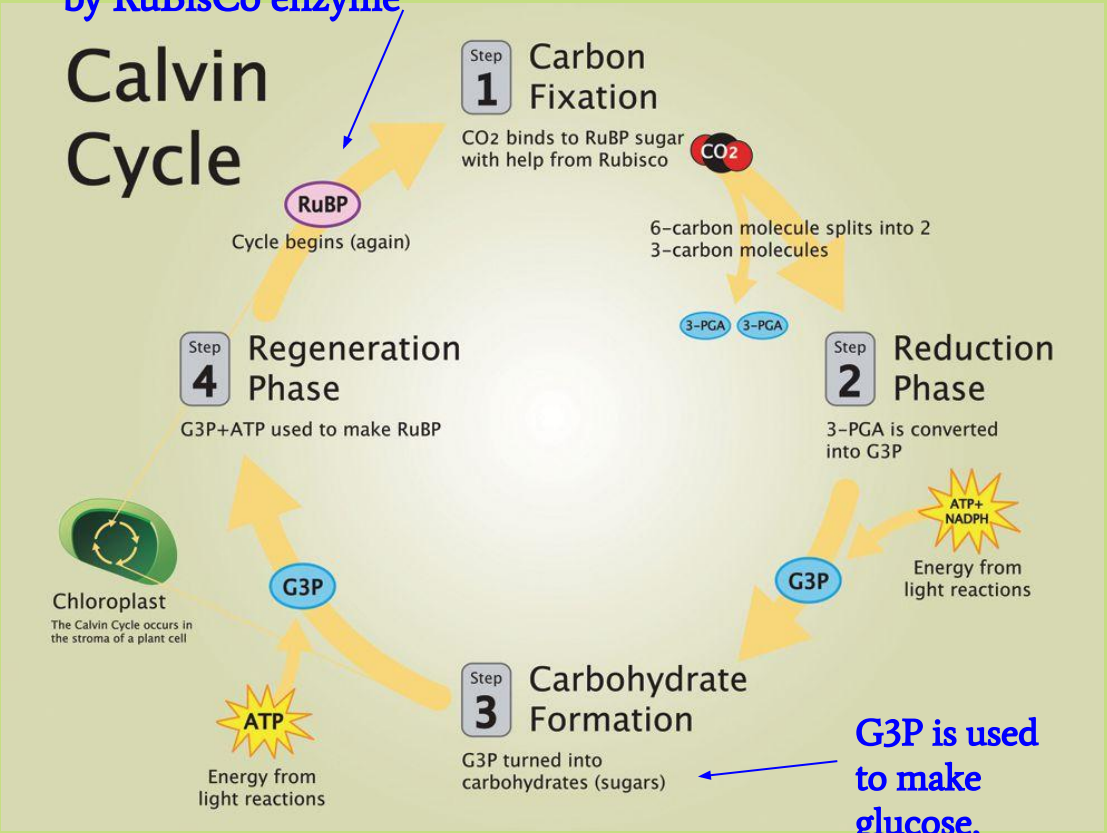


The Calvin Cycle

3 Major phases:

1. Carbon Fixation (**Carboxylation**) of CO2 by RuBisCo
2. Reduction of 3-PGA to G3P
3. Regeneration of RuBP

RuBP is a 5C molecule; gets combined with CO2 by RuBisCo enzyme

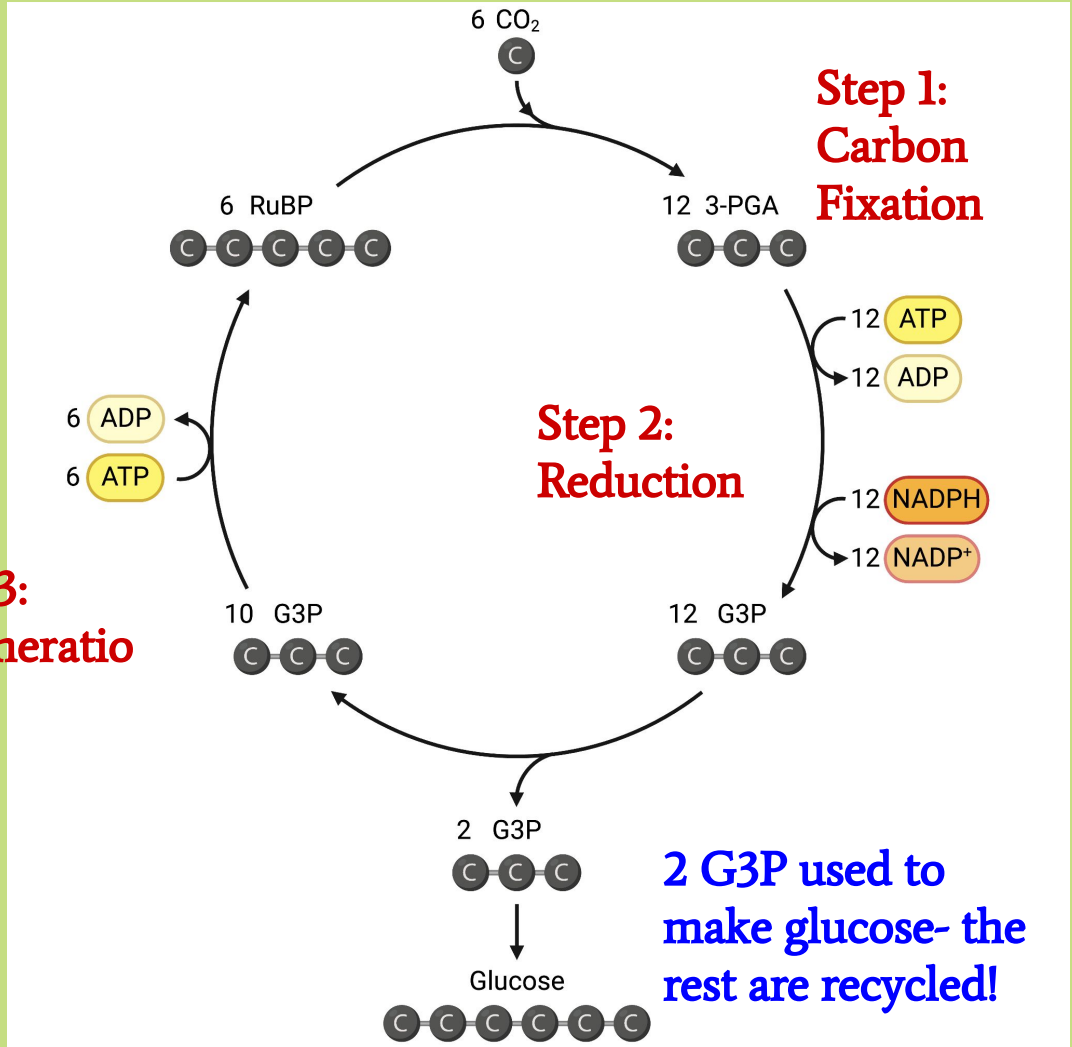


G3P is used to make glucose, starch, cellulose, etc.

Calvin Cycle Video

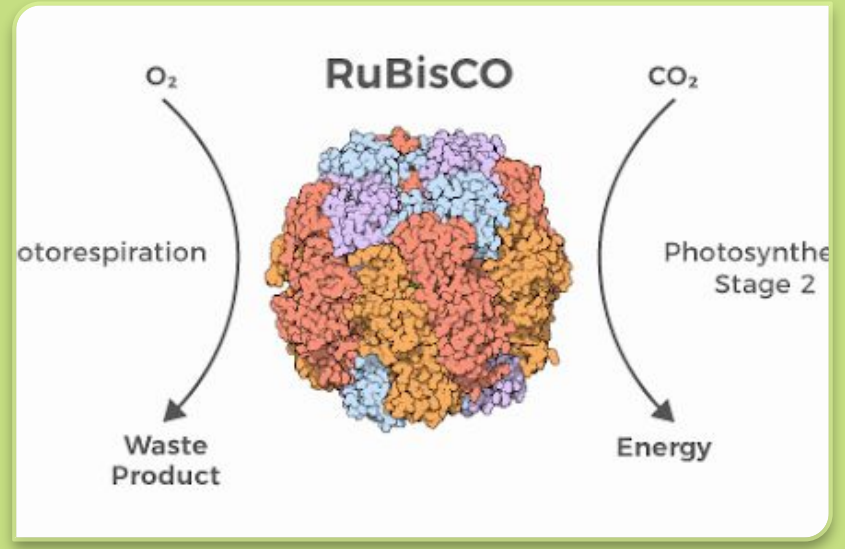


The Calvin Cycle



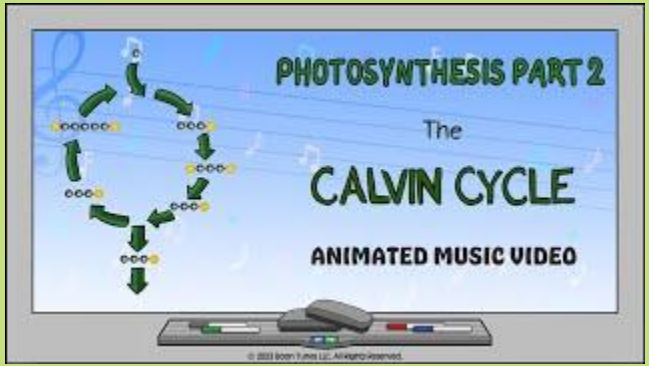
Carbon Fixing Enzyme

- **RuBisCo** = ribulose bisphosphate carboxylase
- Enzyme that fixes carbon from the CO_2 in the air
- Most **important and abundant** enzyme on earth makes life out of air!



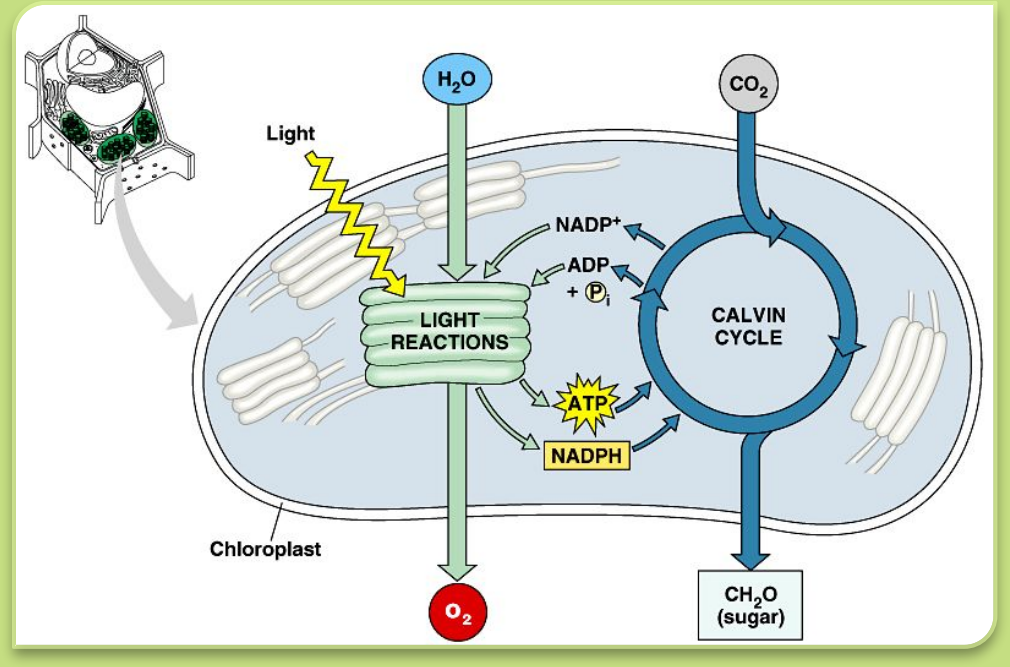
Calvin Cycle Summary

- During 6 turns of the Calvin cycle:
  - 6 CO_2 → 1 $C_6H_{12}O_6$
  - 18 ATP and 12 NADPH are used (where from?)
- Any leftover ATP will be used elsewhere by the cell

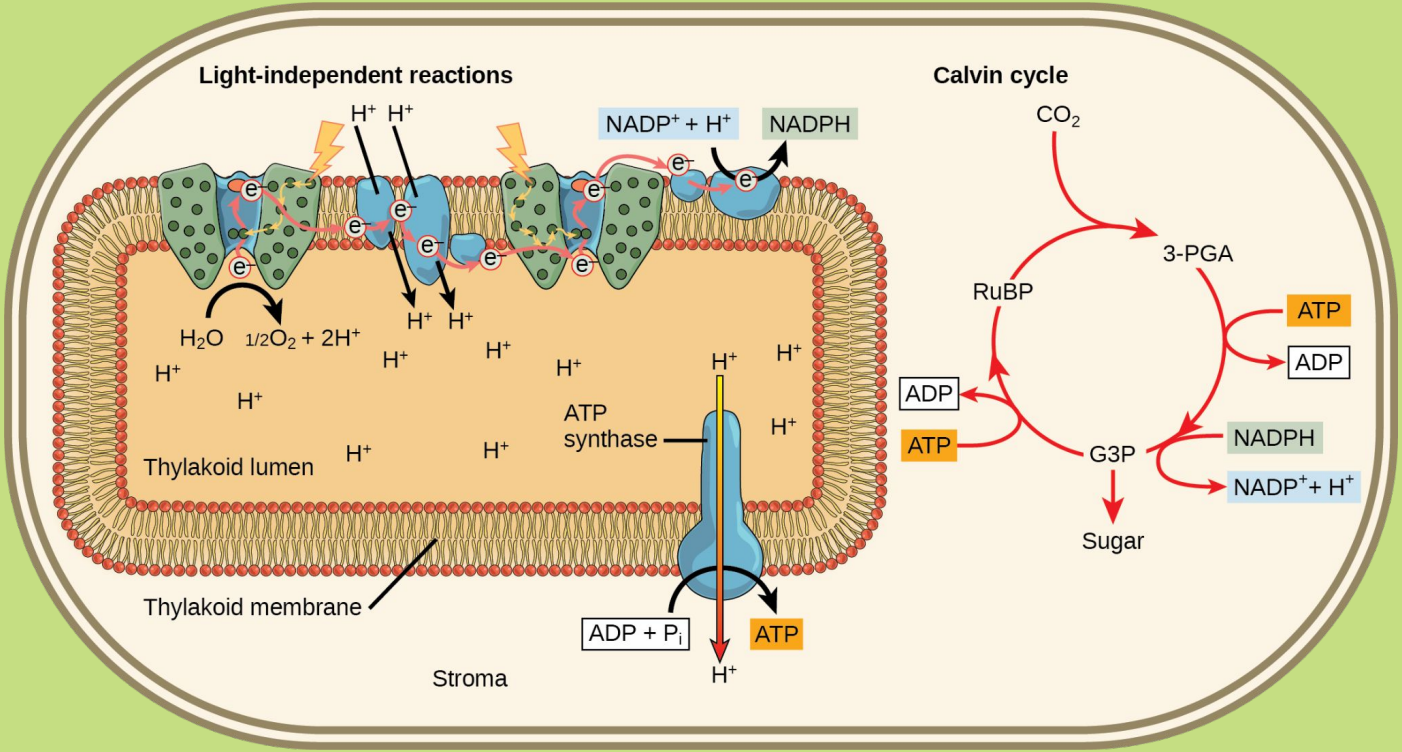


Summary of Photosynthesis

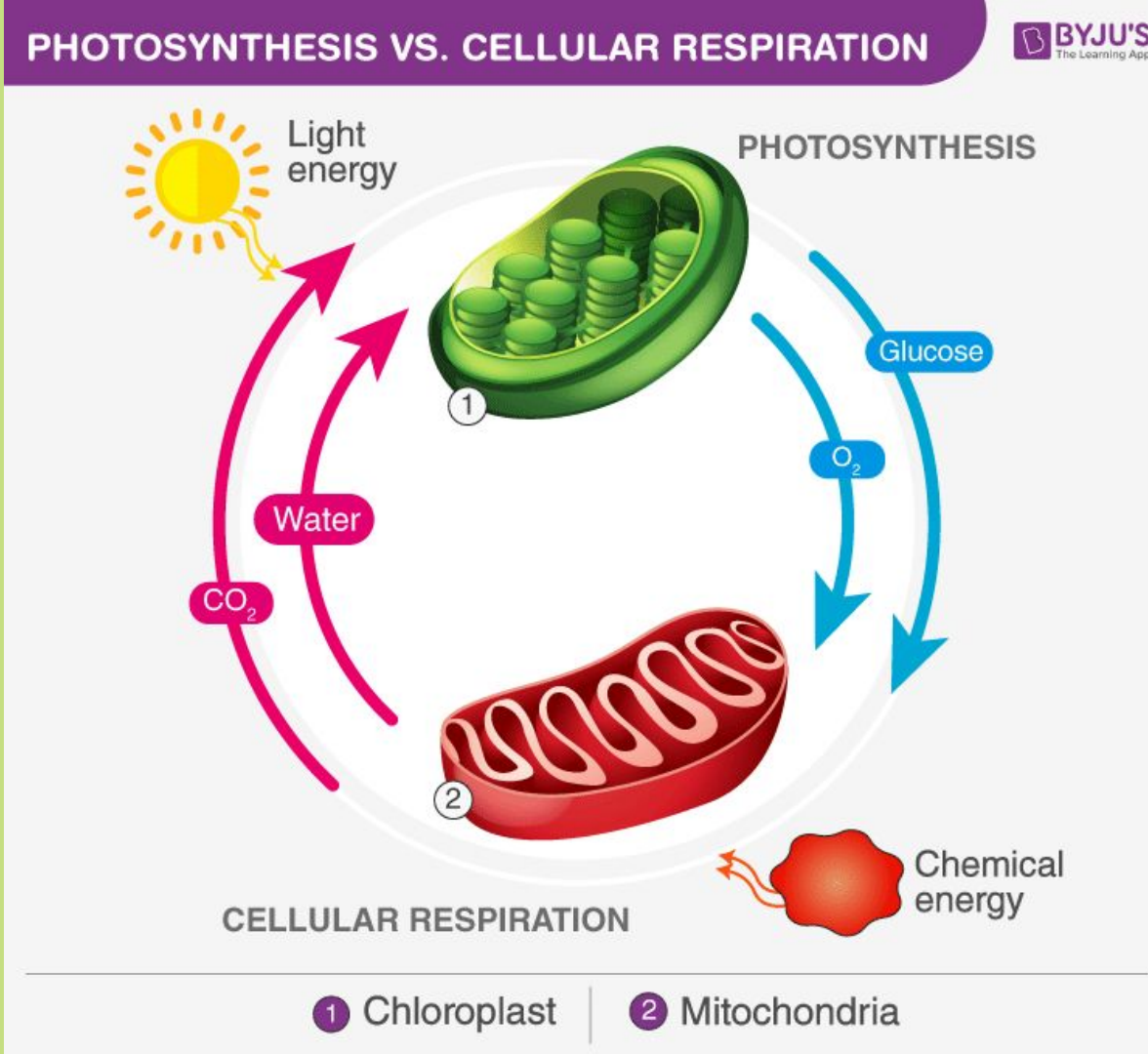
- Light Reactions:
  - Uses H_2O
  - Produces ATP and NADPH
  - Releases O_2 as a byproduct
- Calvin Cycle
  - Uses CO_2
  - Produces $C_6H_{12}O_6$
  - Regenerates ADP and NADP+



Summary of Photosynthesis



The Energy Cycle



Supporting a Biosphere

- On a global scale, photosynthesis is the most important process to life on Earth
- Each year photosynthesis:
  - Captures 121 billion tons of CO_2
  - Synthesizes 275 billion tons of carbohydrates
- Producers and consumers depend on each other to be able to carry out life's processes!



